TCPDUMP & LIBPCAP

LINK-LAYER HEADER TYPES

The table below lists link-layer header types used in pcap and pcap-ng capture files. The `LINKTYPE_` name is the name given to that link-layer header type, and the `LINKTYPE_` value is the numerical value used in capture files. The `DLT_` name is the name corresponding to the value (specific to the packet capture method and device type) returned by **pcap_datalink**(3PCAP); in most cases, as **pcap-linktype**(7) mentions it, the `LINKTYPE_` value and `DLT_` value are the same, but, in some cases, they differ, for reasons of binary compatibility, with the `DLT_` value being different on different platforms.

Note that these values correspond to particular header formats; there might be multiple link-layer header types for a given link-layer protocol, as, for a given link-layer header type, the header for the link-layer protocol might be preceded by a pseudo-header giving additional information, or might be transformed in some way from the way it's specified for that link-layer protocol, e.g. fields in the link-layer protocol header might be removed, added, moved, or modified.

Assigned and Reserved Values

- Values in the range `DLT_USER0`–`DLT_USER15` (147–162, both inclusive) are reserved for private use; if you have some link-layer header type that you want to use within your organization, with the capture files using that link-layer header type not ever be sent outside your organization, you can use one or more of these values. No libpcap release will use these for any purpose, nor will any tcpdump release use them, either.
  Do **NOT** use these in capture files that you expect anybody not using your private versions of capture-file-reading tools to read; in particular, do **NOT** use them in products, otherwise you may find that people won't be able to use tcpdump, or snort, or Wireshark, or… to read the capture files from your firewall/intrusion detection/traffic monitoring/etc. appliance, or whatever product uses that link-layer header type value, and you may also find that the developers of those applications will not accept patches to let them read those files.

Please also be aware that there is an IETF process at: pcaplinktype which will replace all above processes with an IANA.org process.
Also, do not use them if somebody might send you a capture using them for *their* private type and tools using them for *your* private type would have to read them.

- All other values listed in the table below are assigned.
- Any values not listed in the table below are reserved for future use and can be assigned subject to due process.

How To Assign a New Value

Please remember that initial prototyping and experimentation can and should be done using the `DLT_USERx` private use values. If you need a value for a particular set of link-layer headers, you must request one; to do so:

- Subscribe to the mailing list and send a message with the request.
  Please include either an indication of the standard that describes the link-layer headers, a link to a **permanent** and **public** page describing the link-layer headers, or a detailed description of the link-layer headers. If the headers do not exactly match the description in a standard or standards—for example, if fields are added, removed, or reordered, or have their size, endianness or format changed—please describe all differences in detail.
- Prepare libpcap changes for the new value:
  - In `pcap/dlt.h` add the new `DLT_` value and the associated comments, increment `DLT_MATCHING_MAX`.
  - In `pcap-common.c` add the new `LINKTYPE_` value (omit the comments if the previous step explains everything sufficiently well), increment `LINKTYPE_MATCHING_MAX`.
  - In `pcap.c` add the new value to `dlt_choices[]`.
  - Check everything and open a pull request with the changes.
- Prepare web site changes for the new value:
  - If there is no other specification readily available for the new header type, create a new `htmlsrc/linktypes/LINKTYPE_xxxxx.html` file and document the new header structure.

  - In `htmlsrc/linktypes.html` (this file) add the new value to the bottom of the table.
  - Generate complete HTML files from the file(s) above as explained here.
  - Check everything and open a pull request with the changes.
- Have patience.
  Be ready to answer questions and to address any technical issues that emerge during the review. This sometimes takes longer than expected, but it is important to get every detail right because after the assignment the new header structure should be set in stone.

Warning

Do **NOT** add new values to this list except as explained above. Otherwise, you run the risk of using a value that's already being used for some other purpose, and of having tools that read libpcap-format captures not being able to handle captures with your new link-layer header type value, with no hope that they will ever be changed to do so (as that would destroy their ability to read captures using that value for that other purpose).

Do **NOT** use any of these link-layer header type values unless your link-layer headers **EXACTLY** match the specification. If they do not exactly match, request a new link-layer header type for them.

Here are all the link-layer header types, their corresponding linktype values, DLT names, and descriptions:

1. LINKTYPE_NULL (0): DLT_NULL - BSD loopback encapsulation.
2. LINKTYPE_ETHERNET (1): DLT_EN10MB - IEEE 802.3 Ethernet (10Mb, 100Mb, 1000Mb, and up).
3. LINKTYPE_EXP_ETHERNET (2): DLT_EN3MB - Experimental Ethernet (3Mb).
4. LINKTYPE_AX25 (3): DLT_AX25 - AX.25 layer 2 packets.
5. LINKTYPE_PRONET (4): DLT_PRONET - Proteon ProNET Token Ring.
6. LINKTYPE_CHAOS (5): DLT_CHAOS - MIT Chaosnet.
7. LINKTYPE_IEEE802_5 (6): DLT_IEEE802 - IEEE 802.5 Token Ring.

8. LINKTYPE_ARCNET_BSD (7): DLT_ARCNET - ARCNET Data Packets with BSD encapsulation.
9. LINKTYPE_SLIP (8): DLT_SLIP - SLIP, with a header giving packet direction.
10. LINKTYPE_PPP (9): DLT_PPP - PPP.
11. LINKTYPE_FDDI (10): DLT_FDDI - FDDI, as specified by ANSI INCITS 239-1994.
12. LINKTYPE_REDBACK_SMARTEDGE (32): DLT_REDBACK_SMARTEDGE - Redback SmartEdge 400/800.
13. LINK (50): DLT_PPP_SERIAL - PPP in HDLC-like framing.
14. LINKTYPE_PPP_ETHER (51): DLT_PPP_ETHER - PPPoE session packets.
15. LINKTYPE_SYMANTEC_FIREWALL (99): DLT_SYMANTEC_FIREWALL - Symantec Enterprise (ex-Axent Raptor) firewall.
16. LINKTYPE_ATM_RFC1483 (100): DLT_ATM_RFC1483 - LLC/SNAP-encapsulated ATM.
17. LINKTYPE_RAW (101): DLT_RAW - IPv4 or IPv6 packets with no link-layer header.
18. LINKTYPE_C_HDLC (104): DLT_C_HDLC - Cisco PPP with HDLC framing.
19. LINKTYPE_IEEE802_11 (105): DLT_IEEE802_11 - IEEE 802.11 wireless LAN.
20. LINKTYPE_ATM_CLIP (106): DLT_ATM_CLIP - Linux Classical IP over ATM.
21. LINKTYPE_FRELAY (107): DLT_FRELAY - Frame Relay LAPF.
22. LINKTYPE_LOOP (108): DLT_LOOP - OpenBSD loopback encapsulation.
23. LINKTYPE_ENC (109): DLT_ENC - Encapsulated packets for IPsec.
24. LINKTYPE_NETBSD_HDLC (112): DLT_HDLC - Cisco HDLC.
25. LINKTYPE_LINUX_SLL (113): DLT_LINUX_SLL - Linux "cooked" capture encapsulation.
26. LINKTYPE_LTALK (114): DLT_LTALK - Apple LocalTalk packets.
27. LINKTYPE_ECONET (115): DLT_ECONET - Acorn Econet.
28. LINKTYPE_IPFILTER (116): DLT_IPFILTER - OpenBSD ipfilter.
29. LINKTYPE_PFLOG (117): DLT_PFLOG - OpenBSD pflog.

30. LINKTYPE_CISCO_IOS (118): DLT_CISCO_IOS - Cisco internal use.
31. LINKTYPE_IEEE802_11_PRISM (119): DLT_PRISM_HEADER - Prism monitor mode information.
32. LINKTYPE_AIRONET_HEADER (120): DLT_AIRONET_HEADER - Aironet 802.11 cards.
33. LINKTYPE_IP_OVER_FC (122): DLT_IP_OVER_FC - IP and ATM over Fibre Channel.
34. LINKTYPE_SUNATM (123): DLT_SUNATM - ATM traffic captured from a SunATM device.
35. LINKTYPE_RIO (124): DLT_RIO - RapidIO.
36. LINKTYPE_PCI_EXP (125): DLT_PCI_EXP - PCI Express.
37. LINKTYPE_AURORA (126): DLT_AURORA - Xilinx Aurora.
38. LINKTYPE_IEEE802_11_RADIOTAP (127): DLT_IEEE802_11_RADIO - Radiotap link-layer information.
39. LINKTYPE_TZSP (128): Tazmen Sniffer Protocol (TZSP) is a generic encapsulation for any other link type, which includes a means to include meta-information with the packet, e.g. signal strength and channel for 802.11 packets, corresponding to DLT_TZSP.
40. LINKTYPE_ARCNET_LINUX (129): ARCnet Data Packets with Linux encapsulation, corresponding to DLT_ARCNET_LINUX.
41. LINKTYPE_JUNIPER_MLPPP (130): Juniper Networks private data link type, corresponding to DLT_JUNIPER_MLPPP.
42. LINKTYPE_JUNIPER_MLFR (131): Juniper Networks private data link type, corresponding to DLT_JUNIPER_MLFR.
43. LINKTYPE_JUNIPER_ES (132): Juniper Networks private data link type, corresponding to DLT_JUNIPER_ES.
44. LINKTYPE_JUNIPER_GGSN (133): Juniper Networks private data link type, corresponding to DLT_JUNIPER_GGSN.

45. LINKTYPE_JUNIPER_MFR (134): Juniper Networks private data link type, corresponding to DLT_JUNIPER_MFR.
46. LINKTYPE_JUNIPER_ATM2 (135): Juniper Networks private data link type, corresponding to DLT_JUNIPER_ATM2.
47. LINKTYPE_JUNIPER_SERVICES (136): Juniper Networks private data link type, corresponding to DLT_JUNIPER_SERVICES.
48. LINKTYPE_JUNIPER_ATM1 (137): Juniper Networks private data link type, corresponding to DLT_JUNIPER_ATM1.
49. LINKTYPE_APPLE_IP_OVER_IEEE1394 (138): Apple IP-over-IEEE 1394 cooked header, corresponding to DLT_APPLE_IP_OVER_IEEE1394.
50. LINKTYPE_MTP2_WITH_PHDR (139): SS7 MTP2 packets, with a pseudo-header, corresponding to DLT_MTP2_WITH_PHDR.
51. LINKTYPE_MTP2 (140): SS7 MTP2 packets, corresponding to DLT_MTP2.
52. LINKTYPE_MTP3 (141): SS7 MTP3 packets, corresponding to DLT_MTP3.
53. LINKTYPE_SCCP (142): SS7 SCCP packets, corresponding to DLT_SCCP.
54. LINKTYPE_DOCSIS (143): DOCSIS MAC frames, as described by the DOCSIS 4.0 MAC and Upper Layer Protocols Interface Specification or earlier specifications for MAC frames, corresponding to DLT_DOCSIS.
55. LINKTYPE_LINUX_IRDA (144): Linux-IrDA packets, corresponding to DLT_LINUX_IRDA.
56. LINKTYPE_IBM_SP (145): IBM SP switch, corresponding to DLT_IBM_SP.
57. LINKTYPE_IBM_SN (146): IBM Next Federation switch, corresponding to DLT_IBM_SN.
58. LINKTYPE_USER0–LINKTYPE_USER15 (147-162): Reserved for private use; see above, corresponding to DLT_USER0–DLT_USER15.
59. LINKTYPE_IEEE802_11_AVS (163): AVS monitor mode information followed by an 802.11 header, corresponding to DLT_IEEE802_11_RADIO_AVS.
60. LINKTYPE_JUNIPER_MONITOR (164): Juniper Networks private data link type, corresponding to DLT_JUNIPER_MONITOR.

61. LINKTYPE_BACNET_MS_TP (165): BACnet MS/TP frames, corresponding to DLT_BACNET_MS_TP.
62. LINKTYPE_PPP_PPPD (166): PPP preceded by a direction octet and an HDLC-like control field, corresponding to DLT_PPP_PPPD.
63. LINKTYPE_JUNIPER_PPPOE (167): Juniper Networks private data link type, corresponding to DLT_JUNIPER_PPPOE.
64. LINKTYPE_JUNIPER_PPPOE_ATM (168): Juniper Networks private data link type, corresponding to DLT_JUNIPER_PPPOE_ATM.
65. LINKTYPE_GPRS_LLC (169): General Packet Radio Service Logical Link Control, as defined by 3GPP TS 04.64, corresponding to DLT_GPRS_LLC.
66. LINKTYPE_GPF_T (170): Transparent-mapped generic framing procedure, as specified by ITU-T Recommendation G.7041/Y.1303, corresponding to DLT_GPF_T.
67. LINKTYPE_GPF_F (171): Frame-mapped generic framing procedure, as specified by ITU-T Recommendation G.7041/Y.1303, corresponding to DLT_GPF_F.
68. LINKTYPE_GCOM_T1E1 (172): Gcom's T1/E1 line monitoring equipment, corresponding to DLT_GCOM_T1E1.
69. LINKTYPE_GCOM_SERIAL (173): Gcom's T1/E1 line monitoring equipment, corresponding to DLT_GCOM_SERIAL.
70. LINKTYPE_JUNIPER_PIC_PEER (174): Juniper Networks private data link type, corresponding to DLT_JUNIPER_PIC_PEER.
71. LINKTYPE_ERF_ETH (175): Endace ERF records of type TYPE_ETH, corresponding to DLT_ERF_ETH.
72. LINKTYPE_ERF_POS (176): Endace ERF records of type TYPE_POS_HDLC, corresponding to DLT_ERF_POS.
73. LINKTYPE_LINUX_LAPD (177): Linux vISDN LAPD frames, corresponding to DLT_LINUX_LAPD.

74. LINKTYPE_JUNIPER_ETHER (178): Juniper Networks private data link type. Ethernet frames prepended with meta-information, corresponding to DLT_JUNIPER_ETHER.
75. LINKTYPE_JUNIPER_PPP (179): Juniper Networks private data link type. PPP frames prepended with meta-information, corresponding to DLT_JUNIPER_PPP.
76. LINKTYPE_JUNIPER_FRELAY (180): Juniper Networks private data link type. Frame Relay frames prepended with meta-information, corresponding to DLT_JUNIPER_FRELAY.
77. LINKTYPE_JUNIPER_CHDLC (181): Juniper Networks private data link type. C-HDLC frames prepended with meta-information, corresponding to DLT_JUNIPER_CHDLC.
78. LINKTYPE_MFR (182): FRF.16.1 Multi-Link Frame Relay frames, corresponding to DLT_MFR.
79. LINKTYPE_JUNIPER_VP (183): Juniper Networks private data link type, corresponding to DLT_JUNIPER_VP.
80. LINKTYPE_A429 (184): ARINC 429 frames. Every frame contains a 32-bit A429 word, in little-endian format, corresponding to DLT_A429.
81. LINKTYPE_A653_ICM (185): ARINC 653 interpartition communication messages. Please refer to the A653-1 standard for more information, corresponding to DLT_A653_ICM.
82. LINKTYPE_USB_FREEBSD (186): USB with FreeBSD header, corresponding to DLT_USB_FREEBSD.
83. LINKTYPE_BLUETOOTH_HCI_H4 (187): Bluetooth HCI UART Transport Layer packets, corresponding to DLT_BLUETOOTH_HCI_H4.
84. LINKTYPE_IEEE802_16_MAC_CPS (188): IEEE 802.16 MAC Common Part Sublayer, corresponding to DLT_IEEE802_16_MAC_CPS.
85. LINKTYPE_USB_LINUX (189): USB packets, beginning with a Linux USB header, corresponding to DLT_USB_LINUX.

86. LINKTYPE_CAN20B (190): Controller Area Network (CAN) v. 2.0B, corresponding to DLT_CAN20B.
87. LINKTYPE_IEEE802_15_4_LINUX (191): IEEE 802.15.4, with address fields padded, as is done by Linux drivers, corresponding to DLT_IEEE802_15_4_LINUX.
88. LINKTYPE_PPI (192): Per-Packet Information header preceding packet data, corresponding to DLT_PPI.
89. LINKTYPE_IEEE802_16_MAC_CPS_RADIO (193): IEEE 802.16 MAC Common Part Sublayer plus radiotap header, corresponding to DLT_IEEE802_16_MAC_CPS_RADIO.
90. LINKTYPE_JUNIPER_ISM (194): Juniper Networks private data link type, corresponding to DLT_JUNIPER_ISM.
91. LINKTYPE_IEEE802_15_4_WITHFCS (195): IEEE 802.15.4 packets with FCS, corresponding to DLT_IEEE802_15_4_WITHFCS.
92. LINKTYPE_SITA (196): Various link-layer types, with a pseudo-header, for SITA, corresponding to DLT_SITA.
93. LINKTYPE_ERF (197): Endace ERF records, corresponding to DLT_ERF.
94. LINKTYPE_RAIF1 (198): Special header prepended to Ethernet packets when capturing from a u10 Networks board, corresponding to DLT_RAIF1.
95. LINKTYPE_IPMB_KONTRON (199): IPMB packet for IPMI, beginning with a 2-byte header, followed by the I2C slave address, followed by the netFn and LUN, etc…, corresponding to DLT_IPMB_KONTRON.
96. LINKTYPE_JUNIPER_ST (200): Juniper Networks private data link type, corresponding to DLT_JUNIPER_ST.
97. LINKTYPE_BLUETOOTH_HCI_H4_WITH_PHDR (201): Bluetooth HCI UART Transport Layer packets with a direction pseudo-header, corresponding to DLT_BLUETOOTH_HCI_H4_WITH_PHDR.
98. LINKTYPE_AX25_KISS (202): KISS frames between a host and an AX.25 TNC, corresponding to DLT_AX25_KISS.

99. LINKTYPE_LAPD (203): Q.921 LAPD frames, corresponding to DLT_LAPD.
100. LINKTYPE_PPP_WITH_DIR (204): PPP, with a direction header, corresponding to DLT_PPP_WITH_DIR.
101. LINKTYPE_C_HDLC_WITH_DIR (205): Cisco PPP with HDLC framing, with a direction header, corresponding to DLT_C_HDLC_WITH_DIR.
102. LINKTYPE_FRELAY_WITH_DIR (206): Frame Relay LAPF, with a direction header, corresponding to DLT_FRELAY_WITH_DIR.
103. LINKTYPE_LAPB_WITH_DIR (207): X.25 LAPB, with a direction header, corresponding to DLT_LAPB_WITH_DIR.
104. LINKTYPE_IPMB_LINUX (209): Legacy names (do not use) for Linux I2C below, corresponding to DLT_IPMB_LINUX or DLT_I2C_LINUX.
105. LINKTYPE_I2C_LINUX (209): Linux I2C packets, corresponding to DLT_I2C_LINUX.
106. LINKTYPE_FLEXRAY (210): FlexRay automotive bus frames or symbols, preceded by a pseudo-header, corresponding to DLT_FLEXRAY.
107. LINKTYPE_MOST (211): Media Oriented Systems Transport (MOST) bus for multimedia transport, corresponding to DLT_MOST.
108. LINKTYPE_LIN (212): Local Interconnect Network (LIN) automotive bus, with a metadata header, corresponding to DLT_LIN.
109. LINKTYPE_X2E_SERIAL (213): X2E-private data link type used for serial line capture, corresponding to DLT_X2E_SERIAL.
110. LINKTYPE_X2E_XORAYA (214): X2E-private data link type used for the Xoraya data logger family, corresponding to DLT_X2E_XORAYA.
111. LINKTYPE_IEEE802_15_4_NONASK_PHY (215): IEEE 802.15.4 packets with PHY header, corresponding to DLT_IEEE802_15_4_NONASK_PHY.
112. LINKTYPE_LINUX_EVDEV (216): Linux evdev events from /dev/input/eventN devices, corresponding to DLT_LINUX_EVDEV.
113. LINKTYPE_GSMTAP_UM (217): GSM Um interface, preceded by a "gsmtap" header, corresponding to DLT_GSMTAP_UM.

114. LINKTYPE_GSMTAP_ABIS (218): GSM Abis interface, preceded by a "gsmtap" header, corresponding to DLT_GSMTAP_ABIS.
115. LINKTYPE_MPLS (219): MPLS, with an MPLS label as the link-layer header, corresponding to DLT_MPLS.
116. LINKTYPE_USB_LINUX_MMAPPED (220): USB packets, beginning with an extended Linux USB header, corresponding to DLT_USB_LINUX_MMAPPED.
117. LINKTYPE_DECT (221): DECT packets, with a pseudo-header, corresponding to DLT_DECT.
118. LINKTYPE_AOS (222): AOS Space Data Link Protocol, corresponding to DLT_AOS.
119. LINKTYPE_WIHART (223): WirelessHART (Highway Addressable Remote Transducer) from the HART Communication Foundation (IEC/PAS 62591), corresponding to DLT_WIHART.
120. LINKTYPE_FC_2 (224): Fibre Channel FC-2 frames, corresponding to DLT_FC_2.
121. LINKTYPE_FC_2_WITH_FRAME_DELIMS (225): Fibre Channel FC-2 frames with SOF and EOF, corresponding to DLT_FC_2_WITH_FRAME_DELIMS.
122. LINKTYPE_IPNET (226): Solaris ipnet, corresponding to DLT_IPNET.
123. LINKTYPE_CAN_SOCKETCAN (227): Controller Area Network (CAN) frames, with a metadata header, corresponding to DLT_CAN_SOCKETCAN.
124. LINKTYPE_IPV4 (228): IPv4 packets with no link-layer header, corresponding to DLT_IPV4.
125. LINKTYPE_IPV6 (229): IPv6 packets with no link-layer header, corresponding to DLT_IPV6.
126. LINKTYPE_IEEE802_15_4_NOFCS (230): IEEE 802.15.4 packets without FCS, corresponding to DLT_IEEE802_15_4_NOFCS.
127. LINKTYPE_DBUS (231): Raw D-Bus messages, corresponding to DLT_DBUS.

128. LINKTYPE_JUNIPER_VS (232): Juniper Networks private data link type, corresponding to DLT_JUNIPER_VS.
129. LINKTYPE_JUNIPER_SRX_E2E (233): Juniper Networks private data link type, corresponding to DLT_JUNIPER_SRX_E2E.
130. LINKTYPE_JUNIPER_FIBRECHANNEL (234): Juniper Networks private data link type, corresponding to DLT_JUNIPER_FIBRECHANNEL.
131. LINKTYPE_DVB_CI (235): DVB-CI messages, with a pseudo-header, corresponding to DLT_DVB_CI.
132. LINKTYPE_MUX27010 (236): Variant of 3GPP TS 27.010 multiplexing protocol, corresponding to DLT_MUX27010.
133. LINKTYPE_STANAG_5066_D_PDU (237): STANAG 5066 D_PDUs, corresponding to DLT_STANAG_5066_D_PDU.
134. LINKTYPE_JUNIPER_ATM_CEMIC (238): Juniper Networks private data link type, corresponding to DLT_JUNIPER_ATM_CEMIC.
135. LINKTYPE_NFLOG (239): Linux netlink NETLINK NFLOG socket log messages, corresponding to DLT_NFLOG.
136. LINKTYPE_NETANALYZER (240): Ethernet frames with Hilscher netANALYZER pseudo-header, corresponding to DLT_NETANALYZER.
137. LINKTYPE_NETANALYZER_TRANSPARENT (241): Ethernet frames with netANALYZER pseudo-header, preamble and SFD, preceded by a Hilscher, corresponding to DLT_NETANALYZER_TRANSPARENT.
138. LINKTYPE_IPOIB (242): IP-over-InfiniBand, corresponding to DLT_IPOIB.
139. LINKTYPE_MPEG_2_TS (243): MPEG-2 Transport Stream transport packets, corresponding to DLT_MPEG_2_TS.
140. LINKTYPE_NG40 (244): Frames from ng4T GmbH's ng40 protocol tester, corresponding to DLT_NG40.
141. LINKTYPE_NFC_LLCP (245): NFC Logical Link Control Protocol frames, with a pseudo-header, corresponding to DLT_NFC_LLCP.

142. LINKTYPE_PFSYNC (246): Packet filter state syncing, corresponding to DLT_PFSYNC.
143. LINKTYPE_INFINIBAND (247): InfiniBand data packets, corresponding to DLT_INFINIBAND.
144. LINKTYPE_SCTP (248): SCTP packets, as defined by RFC 4960, with no lower-level protocols such as IPv4 or IPv6, corresponding to DLT_SCTP.
145. LINKTYPE_USBPCAP (249): USB packets, beginning with a USBPcap header, corresponding to DLT_USBPCAP.
146. LINKTYPE_RTAC_SERIAL (250): Serial-line packets from the Schweitzer Engineering Laboratories "RTAC" product, corresponding to DLT_RTAC_SERIAL.
147. LINKTYPE_BLUETOOTH_LE_LL (251): Bluetooth Low Energy link-layer packets, corresponding to DLT_BLUETOOTH_LE_LL.
148. LINKTYPE_WIRESHARK_UPPER_PDU (252): Upper-protocol layer PDU saves from Wireshark; the actual contents are determined by two tags, one or more of which is stored with each packet, corresponding to DLT_WIRESHARK_UPPER_PDU.
149. LINKTYPE_NETLINK (253): Linux Netlink capture encapsulation, corresponding to DLT_NETLINK.
150. LINKTYPE_BLUETOOTH_LINUX_MONITOR (254): Bluetooth Linux Monitor, corresponding to DLT_BLUETOOTH_LINUX_MONITOR.
151. LINKTYPE_BLUETOOTH_BREDR_BB (255): Bluetooth Basic Rate and Enhanced Data Rate baseband packets, corresponding to DLT_BLUETOOTH_BREDR_BB.
152. LINKTYPE_BLUETOOTH_LE_LL_WITH_PHDR (256): Bluetooth Low Energy link-layer packets, with a pseudo-header, corresponding to DLT_BLUETOOTH_LE_LL_WITH_PHDR.
153. LINKTYPE_PROFIBUS_DL (257): PROFIBUS data link layer packets, corresponding to DLT_PROFIBUS_DL.

154. LINKTYPE_PKTAP (258): Apple PKTAP capture encapsulation, corresponding to DLT_PKTAP.
155. LINKTYPE_EPON (259): Ethernet-over-passive-optical-network packets, including preamble octets, corresponding to DLT_EPON.
156. LINKTYPE_IPMI_HPM_2 (260): IPMI HPM.2 trace packets, corresponding to DLT_IPMI_HPM_2.
157. LINKTYPE_ZWAVE_R1_R2 (261): Z-Wave RF profile R1 and R2 packets, corresponding to DLT_ZWAVE_R1_R2.
158. LINKTYPE_ZWAVE_R3 (262): Z-Wave RF profile R3 packets, corresponding to DLT_ZWAVE_R3.
159. LINKTYPE_WATTSTOPPER_DLM (263): WattStopper Digital Lighting Management (DLM) and Legrand Nitoo Open protocol packets, corresponding to DLT_WATTSTOPPER_DLM.
160. LINKTYPE_ISO_14443 (264): Messages between ISO 14443 contactless smartcards (Proximity Integrated Circuit Card, PICC) and card readers (Proximity Coupling Device, PCD), with the message format specified by the PCAP format for ISO14443 specification, corresponding to DLT_ISO_14443.
161. LINKTYPE_RDS (265): IEC 62106 Radio data system (RDS) groups, corresponding to DLT_RDS.
162. LINKTYPE_USB_DARWIN (266): USB packets captured on a Darwin-based operating system (macOS, etc.), corresponding to DLT_USB_DARWIN.
163. LINKTYPE_OPENFLOW (267): OpenFlow messages with an additional 12-octet header, as used in OpenBSD switch interface monitoring, corresponding to DLT_OPENFLOW.
164. LINKTYPE_SDLC (268): SNA SDLC packets, corresponding to DLT_SDLC.
165. LINKTYPE_TI_LLN_SNIFFER (269): TI LLN sniffer frames, corresponding to DLT_TI_LLN_SNIFFER.
166. LINKTYPE_LORATAP (270): LoRaWan packets with a LoRaTap pseudo-header, corresponding to DLT_LORATAP.

167. LINKTYPE_VSOCK (271): Protocol for communication between host and guest machines in VMware and KVM hypervisors, corresponding to DLT_VSOCK.
168. LINKTYPE_NORDIC_BLE (272): Messages to and from a Nordic Semiconductor nRF Sniffer for Bluetooth LE packets, corresponding to DLT_NORDIC_BLE.
169. LINKTYPE_DOCSIS31_XRA31 (273): DOCSIS packets and bursts, preceded by a pseudo-header giving metadata about the packet, corresponding to DLT_DOCSIS31_XRA31.
170. LINKTYPE_ETHERNET_MPACKET (274): IEEE 802.3 mPackets, corresponding to DLT_ETHERNET_MPACKET.
171. LINKTYPE_DISPLAYPORT_AUX (275): DisplayPort AUX channel monitoring messages, corresponding to DLT_DISPLAYPORT_AUX.
172. LINKTYPE_LINUX_SLL2 (276): Linux "cooked" capture encapsulation v2, corresponding to DLT_LINUX_SLL2.
173. LINKTYPE_SERCOS_MONITOR (277): Sercos Monitor, corresponding to DLT_SERCOS_MONITOR.
174. LINKTYPE_OPENVIZSLA (278): OpenVizsla FPGA-based USB sniffer frames, corresponding to DLT_OPENVIZSLA.
175. LINKTYPE_EBHSCR (279): Elektrobit High Speed Capture and Replay (EBHSCR) format, corresponding to DLT_EBHSCR.
176. LINKTYPE_VPP_DISPATCH (280): Records in traces from the http://fd.io VPP graph dispatch tracer, in the graph dispatcher trace format, corresponding to DLT_VPP_DISPATCH.
177. LINKTYPE_DSA_TAG_BRCM (281): Ethernet frames, with a Broadcom switch tag inserted, corresponding to DLT_DSA_TAG_BRCM.
178. LINKTYPE_DSA_TAG_BRCM_PREPEND (282): Ethernet frames, with a Broadcom switch tag prepended, corresponding to DLT_DSA_TAG_BRCM_PREPEND.

179. LINKTYPE_IEEE802_15_4_TAP (283): IEEE 802.15.4 packets, with a pseudo-header containing TLVs with metadata preceding the 802.15.4 header, corresponding to DLT_IEEE802_15_4_TAP.
180. LINKTYPE_DSA_TAG_DSA (284): Ethernet frames, with a Marvell DSA switch tag inserted, corresponding to DLT_DSA_TAG_DSA.
181. LINKTYPE_DSA_TAG_EDSA (285): Ethernet frames, with a Marvell EDSA switch tag inserted, corresponding to DLT_DSA_TAG_EDSA.
182. LINKTYPE_ELEE (286): Reserved for ELEE lawful intercept protocol, corresponding to DLT_ELEE.
183. LINKTYPE_Z_WAVE_SERIAL (287): Serial frames transmitted between a host and a Z-Wave chip over an RS-232 or USB serial connection, as described in section 5 of the Z-Wave Serial API Host Application Programming Guide, corresponding to DLT_Z_WAVE_SERIAL.
184. LINKTYPE_USB_2_0 (288): USB 2.0, 1.1, or 1.0 packets, corresponding to DLT_USB_2_0.
185. LINKTYPE_ATSC_ALP (289): ATSC Link-Layer Protocol frames, corresponding to DLT_ATSC_ALP.
186. LINKTYPE_ETW (290): Event Tracing for Windows messages, corresponding to DLT_ETW.
187. LINKTYPE_NETANALYZER_NG (291): Reserved for Hilscher Gesellschaft fuer Systemautomation mbH netANALYZER NG hardware and software, corresponding to DLT_NETANALYZER_NG.
188. LINKTYPE_ZBOSS_NCP (292): ZBOSS NCP Serial Protocol, with a pseudo-header, corresponding to DLT_ZBOSS_NCP.
189. LINKTYPE_ZBOSS_NCP (292): ZBOSS NCP Serial Protocol, with a pseudo-header, corresponding to DLT_ZBOSS_NCP.
190. LINKTYPE_USB_2_0_LOW_SPEED (293): Low-Speed USB 2.0, 1.1, or 1.0 packets, corresponding to DLT_USB_2_0_LOW_SPEED.

191. LINKTYPE_USB_2_0_FULL_SPEED (294): Full-Speed USB 2.0, 1.1, or 1.0 packets, corresponding to DLT_USB_2_0_FULL_SPEED.
192. LINKTYPE_USB_2_0_HIGH_SPEED (295): High-Speed USB 2.0 packets, corresponding to DLT_USB_2_0_HIGH_SPEED.
193. LINKTYPE_AUERSWALD_LOG (296): Auerswald Logger Protocol packets, corresponding to DLT_AUERSWALD_LOG.
194. LINKTYPE_ZWAVE_TAP (297): Z-Wave packets, with a metadata header, corresponding to DLT_ZWAVE_TAP.
195. LINKTYPE_SILABS_DEBUG_CHANNEL (298): Silicon Labs debug channel protocol, as described in the specification, corresponding to DLT_SILABS_DEBUG_CHANNEL.
196. LINKTYPE_FIRA_UCI (299): Ultra-wideband (UWB) controller interface protocol (UCI), corresponding to DLT_FIRA_UCI.
197. LINKTYPE_MDB (300): MDB (Multi-Drop Bus) messages, with a pseudo-header, corresponding to DLT_MDB.
198. LINKTYPE_DECT_NR (301): DECT-2020 New Radio (NR) MAC layer, corresponding to DLT_DECT_NR.